

Hafeez Khan

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EDUCATION

- **Florida Institute of Technology** Florida, USA
Master's & Ph.D. in Computer Science (PhD Advisor: Prof. Siddhartha Bhattacharyya) 2023 - 2027 (Expected)
- **Birla Institute of Technology and Science, Pilani** Dubai, UAE
Bachelors of Engineering in Computer Science (Thesis Advisor: Prof. Raja Muthalaghu) 2019 - 2023

RESEARCH EXPERIENCE

Incoming Machine Learning Intern at RTX Technology Research Center (RTRC) Fall 2026

- Topics: Agentic Large Language Models, Post-training, RLHF, RLAI, DPO, Formal Verification, OOD Detection

Systems Engineering Intern at Collins Aerospace Summer 2026
Guide: Janet Liu, Cong Liu Collins Aerospace, USA

- Topics: Path Planning, Representation Learning, Agentic AI, Trustworthy AI, Simulation, Explainable AI
- Core contributor to the **NASA ULI TRANSCEND** project; Developed a **representation learning-based** path planning model for runtime flight rerouting, improving planning efficiency by **86.67%** over the standard A* algorithm.
- Created an end-to-end simulation pipeline for flight rerouting and generated a synthetic dataset of over **100,000 flight scenarios with simulated airspace hazards**.
- Developed explainable AI methods to assess model robustness and provide reliable rerouting recommendations for pilots.

Research Assistant at ASSIST Lab (Florida Tech) Fall 2023 – Present

- **NASA-Funded Research** Fall 2024 – Present
Guide: Natasha Neogi, Sarah Lehman, Siddhartha Bhattacharyya, Philip Chan NASA LaRC & Florida Tech

- Topics: Pre-training, Self-Supervised Learning, Multi-modal Learning, Diffusion Models, Reinforcement Learning
- Developed a self-supervised world model, improving **ImageNet-1k Top-1 accuracy by 2.1%** over baselines with linear probing, and achieving similar gains on semantic segmentation, object and OOD detection benchmarks.
- Created a post-training pipeline for the flight path-planning agent using DPO, reducing safety constraint violations by **12%** while maintaining **91%** planning efficiency.
- Developed a test-time input refinement framework for few-shot segmentation, improving mIoU by up to **36%** across five diverse benchmark datasets, including Pascal-5i and COCO-20i.

- **Microsoft Research (Collaboration)** Spring 2025
Guide: Yash Jain, Vibhav Vineet Microsoft Research, USA

- Topics: Image Generation, Multi-modal Large Language Models, Diffusion Models, Text-to-Image, Optimization
- Developed a closed-loop test-time prompt refinement framework using multimodal LLMs, improving text-to-image alignment by **11.7%** on DALL-E 3, **12.9%** on FLUX, and up to **15.25%** on Stable Diffusion 1.5/2.1/3 models.

- **DARPA-Funded Research** Summer 2025
Guide: Junaid Babar, Isaac Amundson, Siddhartha Bhattacharyya Collins Aerospace, USA

- Topics: Trust and Safety, Model Checking, Formal Verification, Reinforcement Learning, Language Translation
- Developed an automated Soar-to-PRISM translator using ANTLR and Java, enabling probabilistic model checking and formal verification of cognitive agents while exposing safety violations undetectable through testing alone.

- **American Heart Association-Funded Research** Summer 2024
Guide: Venkat Keshav Chivukula UTHealth Houston, USA

- Topics: 3D/4D Reconstruction, Representation Learning, Medical Imaging, Computational Fluid Dynamics (CFD)
- Developed a 3D autoencoder for intraventricular velocity reconstruction, outperforming UNet3D by **10.55%** PSNR.
- Generated synthetic CFD datasets across **eight LVAD patient geometries** to train 3D models.

AWARDS

- **Awarded 1st Place by Meta** in the Computationally Optimal Gaussian Splatting Competition at the IEEE/CVF International Conference on Computer Vision (ICCV) 2025. 2025
- Awarded for **Outstanding Academic Achievement** at Florida Tech. 2025
- **Undergraduate Research Award** for outstanding Bachelors Thesis at BITS Pilani Dubai. 2023
- **Gold Medalist** in IEEE Robotics Competition, 3D Object Detection and Tracking for Road Safety, held in Dubai. 2023

SKILLS

Programming Languages and Web Apps: Python, R, Matlab, C/C++, Java, SQL, Flask, Streamlit

Proficient Deep Learning and Vision Libraries: PyTorch, TensorFlow, HuggingFace, Transformers, OpenCV

Cloud Platforms & Services: Azure (ML Studio, DevOps), AWS (S3, SageMaker), GCP (Google Maps, Cloud Storage).

SELECTED PUBLICATIONS

* equal contribution

It Is Not the Model, It Is the Input: Test-Time Input-Space Refinement for Few-Shot Segmentation

M.A. Hafeez Khan, Parth Ganeriwala, Sarah M. Lehman, Siddhartha Bhattacharyya, Amy Alvarez, Natasha Neogi
Under Review at an A* Conference, 2026

Adapt, But Don't Forget: Fine-Tuning and Contrastive Routing for Lane Detection under Distribution Shift [\[PDF\]](#)

M.A. Hafeez Khan, Parth Ganeriwala, Sarah M. Lehman, Siddhartha Bhattacharyya, Amy Alvarez, Natasha Neogi
[**Oral Presentation**] International Conference on Computer Vision 2COOL Workshop (ICCV), 2025

AssistTaxi-v2: A Scalable Dataset for Taxiway/Runway Scene Understanding in Diverse Conditions [\[PDF\]](#)

Parth Ganeriwala, M.A. Hafeez Khan, Amy Alvarez, Siddhartha Bhattacharyya, Natasha Neogi, Sarah M. Lehman
American Institute of Aeronautics and Astronautics (AIAA) SCITECH Forum, 2026

Test-time Prompt Refinement for Text-to-Image Models [\[PDF\]](#)

M.A. Hafeez Khan*, Yash Jain*, Siddhartha Bhattacharyya, Vibhav Vineet
[**Invited Talk**] International Conference on Computer Vision MARS2 Workshop (ICCV), 2025

Few-Shot Classification and Anatomical Localization of Tissues in SPECT Imaging [\[PDF\]](#)

M.A. Hafeez Khan, Samuel M. Boddepalli, Siddhartha Bhattacharyya, Debasis Mitra
[**Oral Presentation**] Medical Imaging Conference (MIC), 2025

LVADNet3D: A Deep Autoencoder for Reconstructing 3D Intraventricular Flow from Sparse Data [\[PDF\]](#)

M.A. Hafeez Khan, Marcello Mattei, Ben Diaz, Ruth White, Siddhartha Bhattacharyya, Venkat Keshav Chivukula
[**Oral Presentation**] International Conference on Machine Learning and Applications (ICMLA), 2025

NORA: A Nephrology-Oriented Representation Learning Approach Towards CKD Classification [\[PDF\]](#)

M.A. Hafeez Khan, T. Bhattacharyya, O. Khan, Noorah Khan, Alina Khan, M.Q. Khan, Sujoy Hajra
[**Spotlight Paper**] International Conference on Machine Learning and Applications (ICMLA), 2025

ALINA: Advanced Line Identification and Notation Algorithm [\[PDF\]](#)

M.A. Hafeez Khan, Parth Ganeriwala, Siddhartha Bhattacharyya, Natasha Neogi, Raja Muthalagu
IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), 2024

A Hybrid BiLSTM-CNN Approach for Intrusion Detection for IoT Applications [\[PDF\]](#)

M.A. Hafeez Khan, Sapna Sadhwani, Raja Muthalagu, Pranav Pawar, K. Suresh
Scientific Reports, Springer, 2024

Classification of Microstructure Images of Metals Using Transfer Learning [\[PDF\]](#)

M.A. Hafeez Khan, Hrishikesh Sabnis, J. Angel Arul Jothi, J. Kanishkha, A.D. Prasad
International Conference on Modelling and Development of Intelligent Systems (MDIS), 2022

Detection of Cavities from Oral Images using Convolutional Neural Networks [\[PDF\]](#)

M.A. Hafeez Khan, Giri Prasad S., J. Angel Arul Jothi
[**Best Paper Award**] IEEE International Conference on Electrical, Computer and Energy Technologies (ICECET), 2022